

Math 514, F13
Quiz 2

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) According to the weather forecast there is a 40% chance of rain for each of the next five days. The weather on any given day is independent of the weather on the other days. Let X be the number of rainy days for the next five days. Find $E[X]$, $\text{Var}(X)$, and $P(X \leq 2)$.

Let X_i be the random variable such that $X_i = \begin{cases} 1 & \text{rain on the } i^{\text{th}} \text{ day,} \\ 0 & \text{otherwise,} \end{cases}$
then $X = \sum_{i=1}^5 X_i$. We know $E[X_i] = 0.4$ and $\text{Var}(X_i) = 0.4(1-0.4) = 0.24$
 $\Rightarrow E[X] = E\left[\sum_{i=1}^5 X_i\right] = \sum_{i=1}^5 E[X_i] = 5 \times 0.4 = 2$
 $\text{Var}(X) = \text{Var}\left(\sum_{i=1}^5 X_i\right) = \sum_{i=1}^5 \text{Var}(X_i) = 5 \times 0.24 = 1.2$
 $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$
 $= \binom{5}{0} 0.6^5 + \binom{5}{1} 0.4 \times 0.6^4 + \binom{5}{2} 0.4^2 \times 0.6^3$
 $= 0.7776 + 5 \times 0.05184 + 10 \times 0.03456$
 $= 0.68256$

Problem 2. (5 points) If $\text{Cov}(X_i, X_j) = ij$, find the covariance $\text{Cov}(X_1 + X_2 + X_3, X_2 + X_4)$ and the correlation $\rho(X_1, X_2)$.

$$\begin{aligned} \text{Cov}(X_1 + X_2 + X_3, X_2 + X_4) &= \text{Cov}(X_1, X_2) + \text{Cov}(X_1, X_4) + \text{Cov}(X_2, X_2) + \text{Cov}(X_2, X_4) \\ &\quad + \text{Cov}(X_3, X_2) + \text{Cov}(X_3, X_4) \\ &= 1.2 + 1.4 + 2.2 + 2.4 + 3.2 + 3.4 \\ &= 2 + 4 + 4 + 8 + 6 + 12 = 36 \\ \rho(X_1, X_2) &= \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1) \text{Var}(X_2)}} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Cov}(X_1, X_1) \text{Cov}(X_2, X_2)}} \\ &= \frac{1.2}{\sqrt{(1 \cdot 1)(2 \cdot 2)}} = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1 \end{aligned}$$